

Name: _____

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30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. (2 points) Find the first three terms of the sequence defined below, starting with $n = 1$.

$$a_1 = 3 \text{ and } a_n = 2a_{n-1} - 1 \text{ for } n \geq 2$$

2. (8 points) Determine the limit of the sequence or show that the sequence diverges. If it converges, find its limit. Answers with no work will earn no credit.

(a) $a_n = \frac{\sqrt{n}}{\sqrt{3n+1}}$

(b) $a_n = \frac{n}{2^n}$

3. (3 points) Write a general formula for the sequence $\{\frac{1}{3}, \frac{-1}{6}, \frac{1}{9}, \frac{-1}{12}, \frac{1}{15}, \frac{-1}{18}, \dots\}$. A complete answer includes **both** a formula for a_n and clear indication of where the index n starts. (Does n start a zero? one? two?)

4. (10 points) Evaluate the following improper integrals, if possible. If the integral is not convergent, answer divergent. **You must use correct notation and show your work.** Answers without work will earn no credit.

(a) $\int_1^4 \frac{1}{\sqrt{12-3x}} dx$

(b) $\int_3^\infty \frac{6}{1+x^2} dx$

5. (2 points) **Without integrating** determine whether the integral $\int_3^\infty \frac{6}{10x+x^2} dx$ converges or diverges. Justify your conclusion. An answer without any work will earn no credit. (HINT: Use your work in Problem 1, above.)